Customer Shopping Analysis

1. Project Overview:

- Objective: This project analyzes customer shopping habits and transactional data to uncover valuable insights into spending patterns, product preferences, and customer behaviors.
- Scope: An end-to-end data analytics workflow covering data cleaning and exploration (Python), database querying (SQL), interactive dashboard creation (Power BI), and strategic business recommendations.

2. Dataset Summary:

- Dimensions: 3,900 rows and 20 columns.
- Key Attributes: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount ($), Location, Season, Review Rating, Subscription Status, Shipping Type, Payment Method, etc.
- Data Quality: Checked for missing values and consistency. Core transactional fields had zero missing values.
- Statistical Highlights:
    - Average Purchase Amount: $59.76 (ranging from $20 to $100).
    - Customer Age Range: 18 to 70 years (Mean: ~44 years).

- Average Review Rating: 3.75 out of 5.0.

3. Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python.

- Data Loading: Imported the dataset using pandas.
- Initial_Exploration: Used data.info() to check structure and data.describe() for summary statistics.
- Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Popular Products: High-volume items include Blouses, Jewelry, Pants, Shirts, and Dresses.
- Average Spend: Customers typically spend around $60 per transaction across categories.
- Features Engineering:
  - Created age_group column by binning customer ages.
  - Created purchase_frequency_days columns from purchase data.
- Database integration: Connected Python script to MySQL and located the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL:

SQL queries were used to extract key business metrics from the database

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. Discount vs. High Spend – Analyzed customers who used a discount but still spent more than the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100

3. Top Rated Products – Identified the top 5 products with the highest average review ratings.

item_purchased	avgrate
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. Shipping Type Comparison – Compared the average purchase amounts between Standard and Express Shipping.

	shipping_type	avgshipping
▶	Express	60.48
	Standard	58.46

5. Subscription Impact Analysis – Examined whether subscribed customers spend more by comparing average spend and total revenue between subscribers and non-subscribers.

	subscription_status	total_customer	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

6. Discount Utilization by Product – Identified the top 5 products with the highest percentage of purchases utilizing discounts.

	item_purchased	discountbill
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. Customer Segmentation – Segmented customers into New, Returning, and Loyal based on their total number of previous purchases and counted each segment.

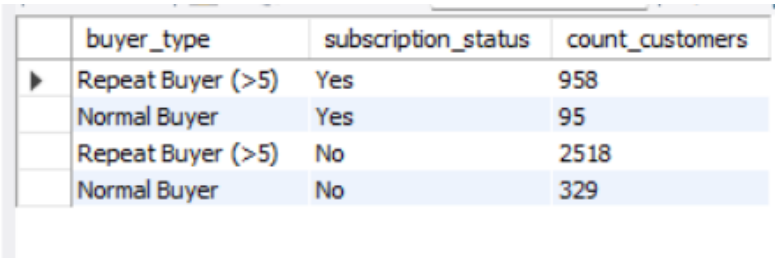
	customer_segment	custid
▶	Loyal	3116
	Returning	701
	New	83

8. Top Products by Category – Determined the top 3 most purchased products within each product category using window functions.



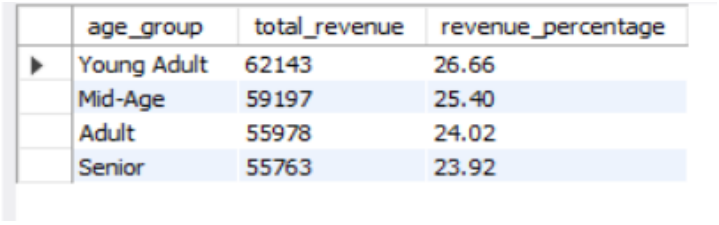
category	item_purchased	item_count
Accessories	Jewelry	171
Accessories	Sunglasses	161
Accessories	Belt	161
Clothing	Blouse	171
Clothing	Pants	171
Clothing	Shirt	169
Footwear	Sandals	160
Footwear	Shoes	150
Footwear	Sneakers	145
Outerwear	Jacket	163
Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Evaluated whether repeat buyers (more than 5 previous purchases) are more likely to subscribe.



buyer_type	subscription_status	count_customers
Repeat Buyer (>5)	Yes	958
Normal Buyer	Yes	95
Repeat Buyer (>5)	No	2518
Normal Buyer	No	329

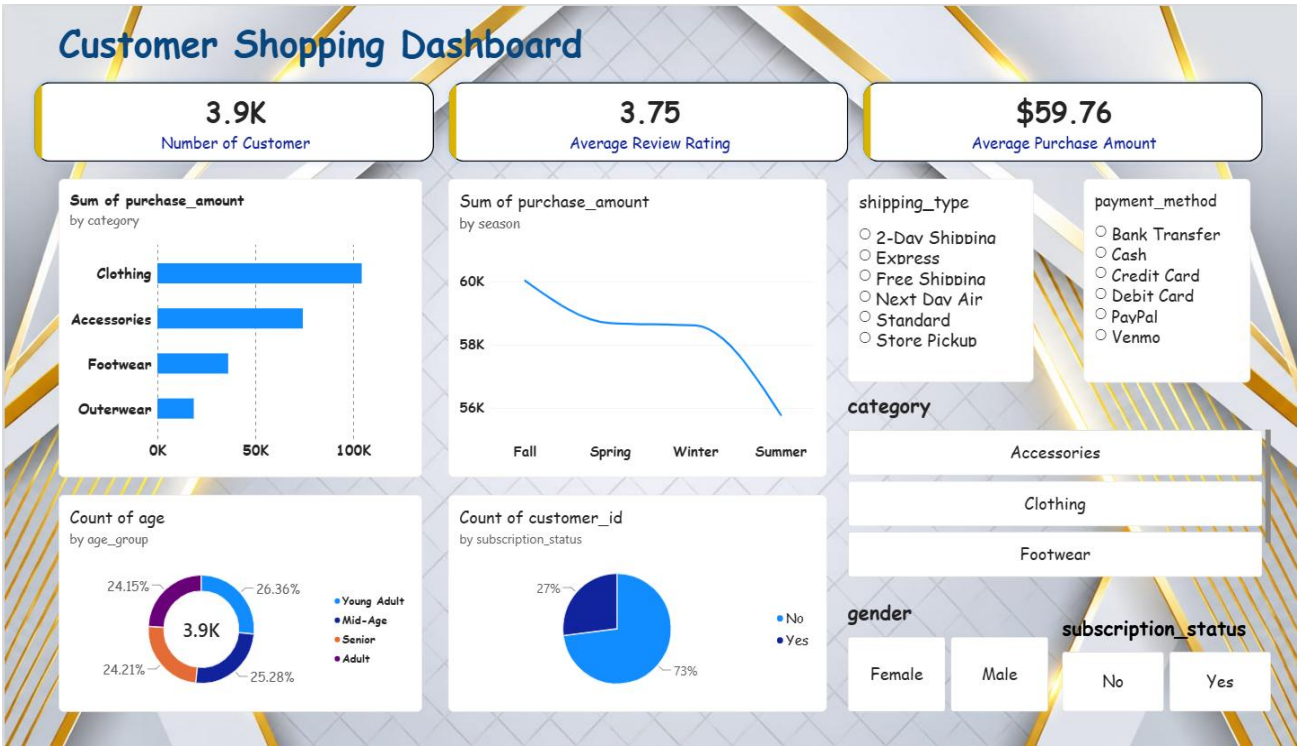
10. Revenue by Age Group – Calculated the revenue contribution and average order value for each age demographic group.



age_group	total_revenue	revenue_percentage
Young Adult	62143	26.66
Mid-Age	59197	25.40
Adult	55978	24.02
Senior	55763	23.92

5. Dashboard in Power BI:

Finally, we built an interactive dashboard in Power BI to presents insights visuall.



6. Business Recommendations

- Boost Subscriptions: Introduce exclusive benefits and loyalty rewards to convert non-subscribers, as subscribers currently represent only ~27% of total orders.
- Inventory Focus: Prioritize inventory management for Clothing and Accessories, which drive the vast majority of total sales.f

- **Payment & Shipping Flexibility: Maintain seamless support across all payment gateways (PayPal, Credit Card, Cash, Debit, Venmo) to reduce cart abandonment**